

Project Report

Project Name: Loan Approval Prediction System

This report summarizes the completed machine-learning capstone system, required tasks, extra features, completion percentage, issues, and conclusion.

1. Project Name

Loan Approval Prediction System

2. Group Members

Member	Role	Related Files
Nabiil	Model Training	train_model.py data/loan_approval.csv models/loan_approval_model.joblib models/model_metrics.csv
Abdalla	PDF Report Generation	reporting.py
Saleh	Prediction Interface and Dashboard	app.py
Jaber	Project Documentation and Setup	README.md requirements.txt .gitignore
Naima	GitHub Manager	GitHub repository final project upload project file organization

3. Required Tasks

Completed required tasks: 13 out of 13

- Import and inspect the loan dataset - Done
- Handle missing values and duplicates - Done
- Encode categorical features - Done
- Perform Exploratory Data Analysis - Done
- Separate features from target variable Loan_Status - Done
- Split data into training and testing sets - Done
- Train at least two classification models - Done
- Evaluate and compare models - Done
- Select and save the best model - Done
- Allow user to enter new applicant information - Done
- Predict Approved / Not Approved - Done
- Display confidence/probability score - Done
- Include functional user interface - Done

4. Extra Features

- Loan approval analytics dashboard

- Feature importance visualization
- Batch prediction using CSV upload
- Downloadable CSV prediction results
- PDF prediction report
- Model comparison table

5. Project Completion Percentage

Project completion: 100%

All required features were completed, and extra features were added.

6. Issues

- Missing values were found in some dataset columns, especially Credit_History, and were solved using an imputation step in the preprocessing pipeline.
- The dataset is small, with 614 records, so we handled this by using a train/test split and comparing multiple models before selecting the best one.
- Very large loan amount inputs can affect predictions because they are outside the normal range of the training dataset, so the dashboard helps users understand the normal loan amount range.
- Streamlit dependencies were required before running the app, and this was solved by adding all required libraries to requirements.txt.

7. Conclusion

The Loan Approval Prediction System was successfully completed. The system uses machine learning to predict whether a loan application is likely to be approved or not approved. Multiple classification models were trained and compared, the best model was saved, and the system was connected to a Streamlit web application.

The final system includes applicant prediction, confidence score, analytics dashboard, feature importance, batch prediction, and PDF report generation. This project demonstrates the full machine-learning workflow from dataset preparation to model deployment in a user-friendly interface.